

M. AHMAD SIDDIQUI

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PROFESSIONAL SUMMARY

Data Scientist and ML Engineer with hands-on experience in deep learning, NLP, computer vision, and edge AI deployment. Built end-to-end systems spanning RAG pipelines, LLM fine-tuning (LoRA/QLoRA), on-device inference with ONNX Runtime and llama.cpp, and real-time computer vision. Skilled at taking projects from research to working implementation.

EXPERIENCE

REZINX

Contractual ML Engineer

Remote

Aug 2025 – Present

- Fine-tuned language models using LoRA for downstream production use cases.
- Built a model and systems monitoring dashboard using Grafana.
- Evaluated multiple embedding models for a job classification task.
- Built a recommendation engine using Superlinked.
- Developed a computer vision hazard detection system to identify fires, knives, and stairs, and assess their proximity to children.

Arch Technologies

Machine Learning Intern

Islamabad, Pakistan (Remote)

May 2025 – June 2025

- Developed machine learning models to analyze large datasets and identify patterns.
- Assisted in data preprocessing and feature engineering for enhanced model accuracy.
- Conducted experiments to evaluate model effectiveness and recommend adjustments.

Ittehad Chemicals

Network Administration Intern

Lahore, Punjab, Pakistan

Jun 2026 – Jul 2026

- Worked on network administration tasks covering microwave WAN links, OSPF routing, Cisco and MikroTik equipment, and Windows Server Active Directory.

EDUCATION

Khwaja Fareed University of Engineering and Information Technology

Bachelor of Science in Data Science | CGPA: 3.35/4.00

Aug 2022 – May 2026

PROJECTS

QuestionPrep: AI-Based Interview Simulation Platform (FYP)

github.com/Copperhorse/QuestionPrep

- Built a full-stack AI interview platform using FastAPI, llama.cpp (LFM-1.2B GGUF), ChromaDB with BGE embeddings, and a three-stage NLI scoring pipeline (fuzzy match → bi-encoder → DeBERTa cross-encoder).
- Implemented real-time stress detection in-browser via a TCN model exported to ONNX Runtime Web, using a custom JavaScript DSP pipeline (FFT, mel filterbank, DCT-II) for feature extraction with no server round-trip.

Hybrid Profanity Detection & Rewriting System (NLP)

github.com/Copperhorse/profanity_distilbert_lora

- Engineered a two-stage inference pipeline combining a LoRA-fine-tuned DistilBERT classifier (95.67% accuracy) with a QLoRA-fine-tuned LFM-1.2B LLM for context-aware rewriting.
- Implemented dynamic adapter switching between classification and generation modes; documented generation collapse as an intrinsic post-classification fine-tuning phenomenon.

Cyber Attack Detection System (Machine Learning)

github.com/Copperhorse/edge_iiot

- Implemented a multi-class network intrusion classifier on the Edge-IIoT dataset, reaching 94.67% accuracy.
- Applied ensemble techniques (stacking/boosting) with feature engineering to improve detection across attack categories.

SKILLS

Programming Languages: Python, SQL

ML & DL Frameworks: PyTorch, scikit-learn, XGBoost, ONNX Runtime, llama.cpp

NLP & LLMs: Transformers (HuggingFace), LoRA, QLoRA, RAG, text classification, embeddings, LLM fine-tuning

Computer Vision: EfficientNet, YOLOv8, transfer learning, object detection, image classification, OpenCV

Data & Backends: FastAPI, ChromaDB, SQLite, Superlinked, vector search, Docker, REST APIs

MLOps & Tools: Grafana, Git, Bash, Jupyter